

3. Research Strategy

A. Significance

Segmenting and grouping texture into regions that are likely to be physically meaningful is a crucial and fundamental visual capability that underlies our ability to interpret and interact with the environment. Although there is a large literature on texture segmentation/grouping, there are no measurements of human texture-discrimination and texture-grouping performance when the textures are randomly sampled on each trial from a wide range of natural textures. Such measurements are necessary for understanding the capabilities and limitations of human perceptual organization mechanisms and for testing formal theories that can explain those capabilities and limitations. Similarly, measuring and formally analyzing the statistical properties of natural textures is necessary for characterizing and understanding the information available to perform texture segmentation/grouping under natural conditions. Finally, developing Bayesian and SSCNN observers is a critical step in the development of principled computational models of human segmentation/grouping performance and of the underlying neurophysiological mechanisms.

B. Innovation

The proposed research is innovative in several ways. (1) It will be the first study to measure human ability to discriminate patches of natural texture in a same-different task when they are randomly sampled from a wide range of texture types. (2) It will be the first study to measure human ability to perceptually organize natural image patches in a rows or columns task when the patches are randomly sampled from a wide range of natural texture types. (3) It will be the first to develop image-computable Bayesian natural statistics models for the general same-different texture discrimination tasks. (4) It will develop a new class of self-supervised deep neural network models that is even more general and may be extensible to other domains of perceptual processing. (5) It will develop perceptual grouping models that include a new testable grouping principle (mutual similarity).

C. Approach

C.1 Background and Preliminary Studies

The human visual system has a remarkable ability to rapidly group and segment natural scenes into regions that most likely contain the same physically meaningful texture. It can do this with arbitrary new scenes, and do it well, even though the resolution of the visual system declines rapidly away from the line of sight. This ability is an essential and powerful factor supporting the visual system's ability to recognize objects and, more generally, correctly interpret natural images. Physically meaningful image regions are often signaled by luminance and chromatic texture that can vary depending on lighting and surface geometry. So, the big question is: How does the visual system rapidly segment the scene into meaningful texture regions, given its foveated retina, the often-

novel textures, and the variations in lighting and geometry? The proposed research focuses on this question for gray scale natural textures.

There is a substantial vision-science literature on texture segmentation and grouping. Much of this literature is directed at the task of segmenting or grouping two neighboring texture regions containing some number of experimenter-designed texture elements where the spacing between elements is approximately the same over the whole stimulus [31,38]. However, texture segmentation and grouping under natural conditions is often quite different. First, natural textures generally cannot be easily described in terms of discrete elements (Fig. 1) and are only approximately statistically ergodic [37]. Second, there are several qualitatively different situations in which textures are grouped and/or segmented. One common situation is when the texture regions are well separated (Fig. 1A). The visual system segments them because they are well separated, but it often simultaneously

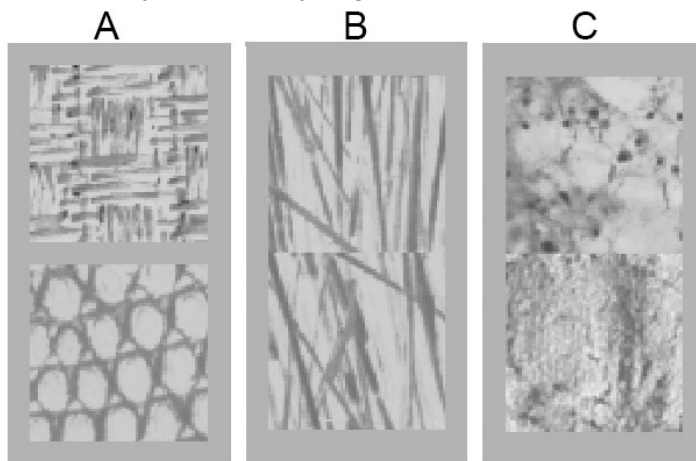


Fig. 1 Texture grouping and segmentation. **A** Separated natural texture patches that are different. Often the visual system must decide whether to group textures that are separated by occluding surfaces. **B** Adjacent random samples of the same natural texture, like what occurs with occlusion or camouflage. **C** Adjacent different natural textures, like what occurs with occlusion or with transition from one type of physical material to another.

groups them together. This occurs, for example, when a large surface with a given texture is partially occluded

by a surface with a different texture. In this situation the useful segmentation/grouping information is in the texture content of the two regions. Another less common but important situation is when adjacent regions contain the same texture content (Fig. 1B). This can occur when one type of object with a given surface texture partially occludes another object of the same type, or when organisms evolve surface texture (camouflage) that mimics the background texture in their natural environment. In this situation, most of the relevant segmentation/grouping information is contained at the boundary between the two regions. Perhaps the most common situation is when regions with different textures are adjacent (Fig. 1C). In this case, the relevant segmentation/grouping information is contained in both the boundary and the texture content. The proposed research in Aims 1 and 2 is directed at getting a better understanding of the mechanisms that underlie human ability to discriminate small natural texture regions under the three situations illustrated in Fig 1. The experiments will measure discrimination performance in a simple single-interval same-different task, where the observer reports whether two texture patches are the same or different. Aim 3 is directed at the mechanisms that group small texture regions into larger structures. The experiments will measure grouping performance in a single-interval task where the observer reports whether the dominant organization in an array of texture patches is rows or columns.

C.1.1 Aim 1: Discrimination of Separated Natural Texture Regions

Most of the early research on texture discrimination was directed at stimuli where the texture regions are adjacent but composed of discrete elements that differ along specific dimensions such as element shape, orientation, size, luminance, contrast, and color [1,7,8,19,20,22,23,31,37,38,44]. Note that because the elements are discrete, sharp edges that coincide with the boundary, like those in Fig. 1B and 1C, are often not formed. Not surprisingly, performance tends to increase as the mean difference between the elements along these dimensions increases and tends to decrease as the level of random jitter of the elements along these dimensions increases [22,23,31,38]. However, the overall pattern of results across studies is complex and there is no model that can handle all the major findings. Nonetheless, the family of filter-rectify-filter (FRF) models [2,3,7,22,23,26] can account for many of the reported phenomena. These image-processing models assume that the input image is initially encoded by multiple channels of linear units. Then a point nonlinearity (e.g., squaring) is applied to the response of each linear unit in the channel, to obtain something like a complex cell response. These rectified responses are then filtered/pooled by another array of linear units giving a map of the something like the average power in each channel over the input image. Finally, these average power maps are combined and thresholded to identify the different texture regions. The heart of the FRF models is the rectification stage, which causes the final output to be roughly invariant over spatial translation of the input texture. In other words, the goal is to produce different responses to different textures while at the same time producing the same response independent of the translations and the other random variations that occur within the same texture. We propose to test some versions of FRF model by generating predictions for our same-different experiments (see C.2.1).

More sophisticated texture-discrimination models have been developed by expanding the set of translation-invariant image properties encoded from the texture samples. For example, Portilla and Simoncelli [29] identified (by trial and error) a large set of translation-invariant properties (e.g., correlations in local energy over position, orientation, scale, spatial frequency, and relative phase) that are sufficiently complete to convincingly simulate a wide range of natural textures. Freeman et al [14] and Lieber et al [25] measured human ability to discriminate such P-S simulated textures from phase-scrambled versions of the same texture in a three-alternative (odd-ball) task where the three texture regions were spatially separated. They found that human performance is correlated with model observers that use the P-S properties (P-S observers). Thus, we also propose to test P-S observers against the results of our same-different experiments. However, the odd-ball task is easier and less natural than the simple same-different task. It is easier because the observer only needs to judge which pairing is the most similar; no absolute judgment of whether a pair is the same texture must be made. It is less natural because in the real world the visual system often must make absolute judgments of whether a pair of textures are the same. It is also a rather unnatural task to compare natural textures with phase scrambled versions of the same texture. Thus, P-S observers may not match human performance as well for the current same-different task.

As mentioned in the Specific Aims and described in the proposed studies (C.2), we have begun developing two other classes of model observer that are tailored for our same-different task, where the stimuli are pairs of relatively small patches of randomly selected natural texture. One class is Bayesian models based on a set of statistical properties of natural textures, where the properties are selected by intuition and trial and error. We then determine/derive the log likelihood ratio that the two natural image patches are from the same vs. different

textures, given the set of properties encoded from each patch. To develop these models we first empirically determine, from a large set of natural textures, a parameterized family of probability distributions that can adequately describe the probability distributions of each property across natural textures (with the parameter values varying across textures). We then run a series of training trials, where on each trial the parameters (typically just one or two) for each property are estimated from the two texture patches and the log likelihood is calculated. From all the training trials we then learn the joint distribution of these log likelihoods and determine the optimal decision bound for making the same-different decision. This approach is like that of the P-S observers, except that we compute the same-different log likelihood for each property using the pre-learned family of distributions. In preliminary studies (see later) we show that just a few properties are sufficient to perform the same-different task with high accuracy. Our goal is to determine which properties best predict human trial-by-trial performance.

The other class of models is self-supervised convolutional neural networks (SSCNNs). Supervised CNNs have been shown to perform very well (reaching human performance) on many naturalistic tasks [24]. Like the FRF models, CNNs consist of convolutional layers (linear filters) separated by layers of point nonlinearities, typically rectified linear (ReLU) functions. CNNs are different in that they consist of more convolutional layers, the convolution layers are followed by one or two fully connected linear-weighting layers, and all the convolutional kernels (linear receptive field shapes) and the fully connected weights are learned by stochastic gradient descent from a very large set of training images. A weakness of supervised CNNs is that they require a very large set of labeled ground-truth images, which is not a realistic learning scenario either during development or during evolution and which often creates the practical problem of obtaining the labeled images.

Nonetheless, if a supervised CNN can learn to use most of the information available to perform a natural task, it may turn out to be a good predictor of human performance. The reasoning is that evolution is likely to drive visual systems to efficiently use most of the sources of information available in natural stimuli for the tasks the organism normally performs. If CNNs and humans both efficiently use most of the available information, then they may perform similarly, even if the implementations are quite different. Furthermore, better predictions of human performance should be possible by including known biological constraints in the CNN (e.g., preprocessing the stimuli with human optics, photoreceptor sampling and/or ganglion cell sampling). In preliminary studies, we trained a supervised CNN for the same-different texture discrimination task and (for a first pass) obtained reasonable performance in the task.

We realized, however, that contrastive self-supervised CNNs are much better suited for the same-different task, which is the approach we propose to take. Contrastive self-supervised CNNs generate category labels from unlabeled images and then train in the same way as a supervised CNN [34]. The same-different task is very well suited for obtaining useful labels directly from the images. Consider a random small patch taken from an arbitrary full color natural image. Patches that are right next to this patch and are simultaneously very similar in color are quite likely to be samples of the same texture. On the other hand, a patch that is distant from this patch and dissimilar in color is quite likely to be a different texture. We have found that this method of labeling produces a large percentage of subjectively plausible labels. Studies of standard CNNs have shown that they are highly robust to labeling errors [30]; even a 50% labeling error rate produces a negligible drop in classification performance on correctly labelled test sets. Our labeling error rate should be substantially smaller, and thus the SSCNNs should perform very well in same-different tasks with correctly labelled test stimuli.

We are enthusiastic about this approach for several reasons. First, this kind of boot-strap learning is evolutionarily plausible. It is easy to imagine that the earliest visual systems used proximity and luminance (assuming a single type of photoreceptor) to segment images into coarse, physically meaningful regions, albeit with limited accuracy. The evolution of additional photoreceptor classes might then have been incorporated into the proximity/luminance segmentation circuits to achieve more accurate segmentation. The evolution of complex spatial mechanisms (e.g., orientation selectivity) might then have been incorporated into the proximity/color segmentation circuits to obtain even more accurate segmentation, and so on. Second, it is easy to have diverse, unlimited training data which will produce robust and stable networks. Such networks might serve as the foundation for networks that solve more complex higher-level tasks. Third, it is easy to train these SSCNNs on arbitrary images. Thus, it will be easy to determine whether the kind of environment (forest, arid, underwater, closeup, distant, urban, indoor) affects the SSCNNs. If it does, then training SSCNNs for the environments of different species might lead to testable predictions of performance in same-different texture discrimination tasks.

(which are tasks many other species can perform). SSCNNs trained on a broad range of natural images should be reasonable observer models to compare with human performance, especially if we include known biological constraints (e.g., optics, photoreceptors, and ganglion cell sampling). We believe that the Bayesian-image-statistics and SSCNN models are not just competitors but are complementary in that the Bayesian models provide deeper insight into the nature of the information available to perform the same-different task and the SSCNN models can help discover other sources of information.

C.1.2 Aim 2: Discrimination of Adjacent Natural Texture Regions

Same-different discrimination of adjacent natural textures is rather different from previous studies of texture discrimination because most previous studies used discrete texture elements that do not create sharp edges coincident with the boundary between the texture regions. We have completed some preliminary studies directed at the case where the adjacent regions are samples of the same naturalist textures (the case of maximum camouflage). We do not yet have preliminary data on the case where the adjacent regions contain samples of the same or different natural texture, but we have completed a related study of detection of occluding targets in natural backgrounds. Below we summarize some of this preliminary work.

C.1.2.1 Discrimination of adjacent regions containing the same natural texture

In one of our preliminary studies, the task was to decide whether a center and surround region are part of one large sample of texture or are two different samples of the same texture (Fig. 2). All the textures, except the “pink” and “filtered” noise, were created from natural textures using the Portilla & Simoncelli [29] algorithm.

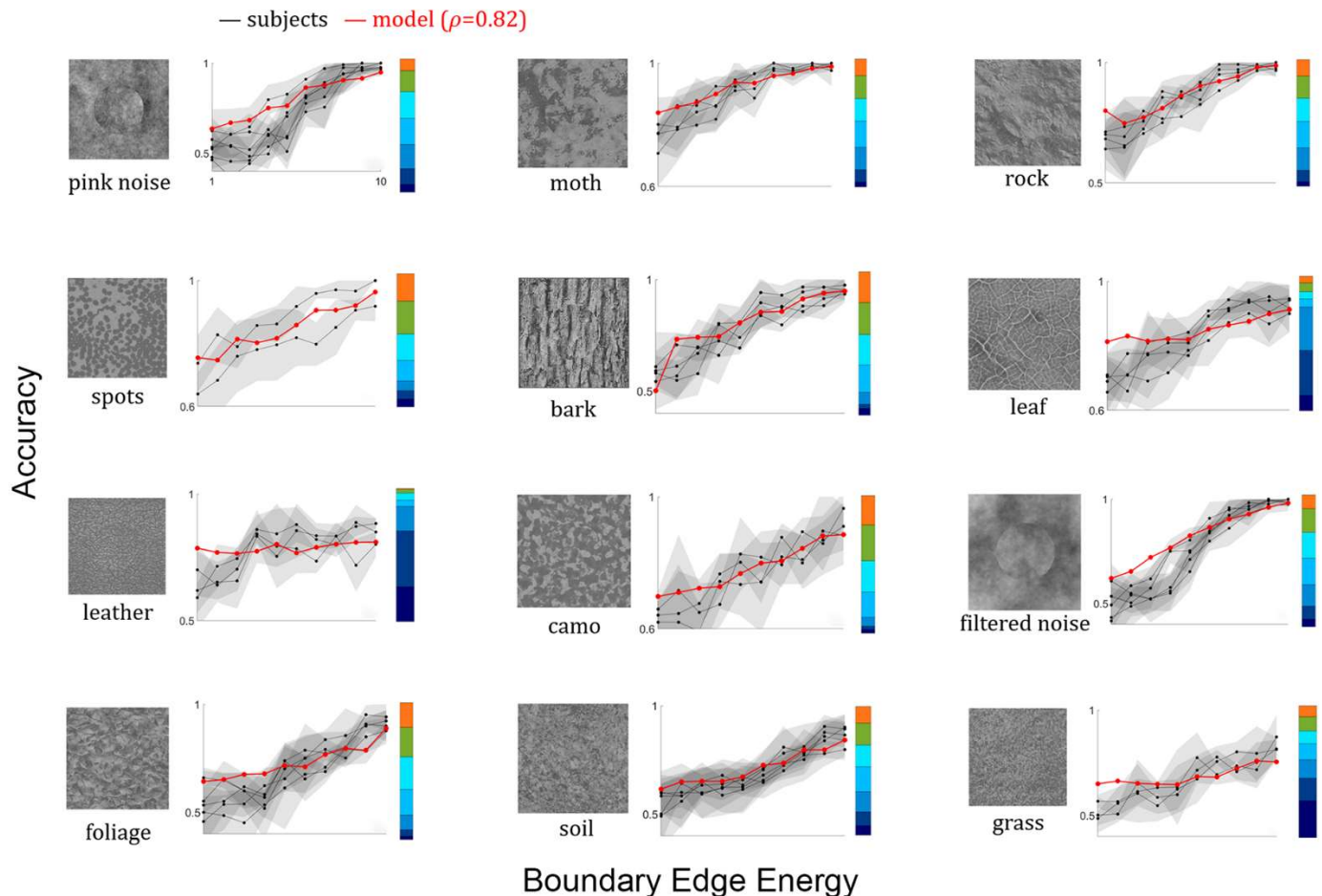


Fig. 2 Discrimination of adjacent regions containing the same texture, for 12 kinds of naturalistic (Portilla & Simoncelli) textures.

The stimuli were presented in a single interval task for a duration of 250 ms (the duration of a typical fixation when searching a visual scene). The psychometric functions for multiple subjects (black points) were obtained by varying the edge energy perpendicular to the boundary. This energy was not directly manipulated. Rather, we simply sorted the randomly generated stimuli into edge energy bins, measured the observer's d' in each bin

and then converted the d' values to bias-corrected accuracy. As can be seen, there are some individual differences, but the results are systematic, and vary across the different kinds of texture. We have begun developing Bayesian observer models using the approach described above (and in the proposed studies). Specifically, we first filtered the images with the human optical transfer function, and then considered seven different properties of contours extracted in the boundary region: the edge-element density, the distribution of contour lengths, and the distribution of edge-element power at five scales. The edge-element densities across all textures are approximately Gaussian, the contour lengths are approximately exponential, and the edge-element powers at all scales are approximately hypo-exponential. Using these distribution shapes we derived, for each property, a close-form expression for the log likelihood of same or different that can be computed on each trial. Finally, we estimated the joint distribution of the seven log likelihoods to construct a single decision variable that can be computed on each trial. This joint multivariate normal distribution was estimated using tools we recently published [9] and made available online (<https://github.com/abhranildas/classify>). The joint distribution and decision variable account for the reliability of, and correlations between, the seven log likelihoods. The red points in Fig. 2 show the predictions of the model, which has only one free parameter: an overall efficiency scalar. The color bars indicate the relative contribution of the seven image properties. This is only a first pass, but it suggests that Bayesian observers based directly on natural scene statistics may be able to account for many aspects of human ability to discriminate maximally camouflaged regions.

C.1.2.2 Discrimination of adjacent regions containing the same or different natural texture

In a recent study [41], we measured human ability to detect occluding targets in natural backgrounds, a task where both boundary and content information are available. Little was known about detection of occluding targets in natural (or other) backgrounds. In addition to the behavioral measurements, we developed a formal theory that includes known factors: human optics, the variation in ganglion-cell sampling across the visual field, and intrinsic position uncertainty. Specifically, we measured detection performance in natural backgrounds for four occluding targets—vertical edge, horizontal edge, bowtie, and spot—all having the same mean luminance (see

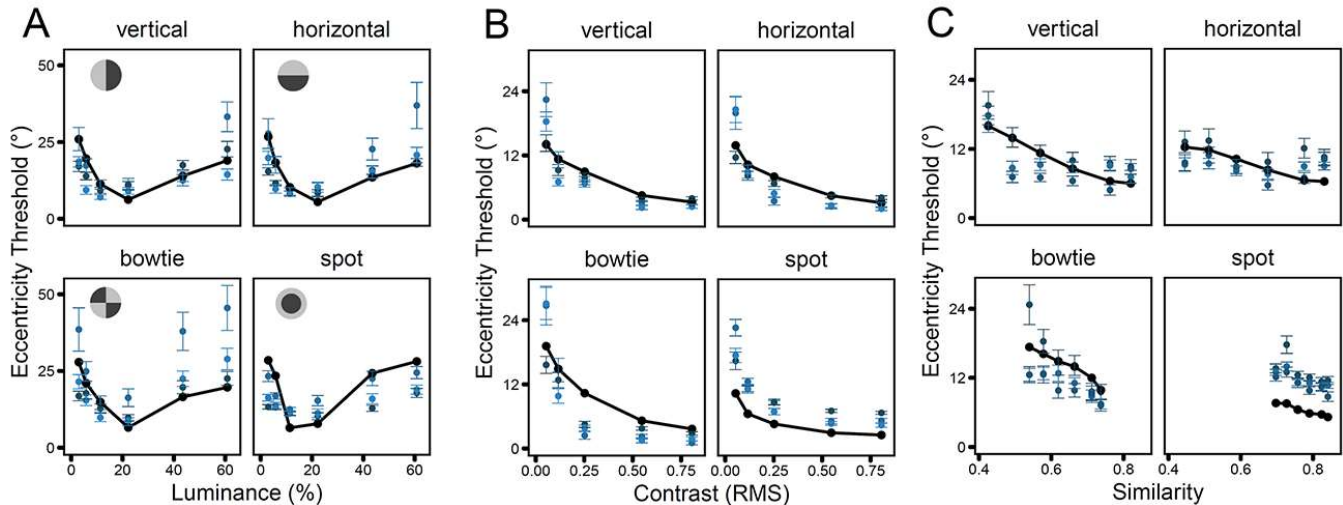


Fig. 3. Eccentricity thresholds as a function of L , C and S , for four different target shapes. There is a separate symbol for each subject's threshold. Solid curves are parameter-free predictions (given human optics, ganglion cell sampling, divisive normalization, and measured intrinsic position uncertainty), except for an overall efficiency scalar. (Walshe & Geisler, JOV, 2020)

Fig. 3A). Because the targets are opaque, they are easily detected in the center of the fovea and decline in detectability with retinal eccentricity. We measured eccentricity psychometric functions in three observers for natural backgrounds that were binned along three dimensions: luminance, contrast, and amplitude spectrum similarity to the target [17]. From each eccentricity psychometric function, we determined an eccentricity threshold: the eccentricity where detectability d' falls to 1.0 (i.e., bias-corrected 69% correct). The symbols in Fig. 3 plot the eccentricity thresholds for each target, along each of the three background dimensions, while the other two dimensions are held at their median values. It is important to note that higher eccentricity thresholds correspond to higher sensitivity—the target is visible further into the periphery. Threshold (sensitivity) decreases as a function of background contrast and similarity for all four targets and is u-shaped as a function of luminance. We developed and tested a principled theory of occluding-target identification that has only one free parameter (an overall efficiency scalar). This near-optimal observer combines three image cues (pattern template,

silhouette template, and edge energy), and four well-known properties of the human visual system (optical blur [43], blurring and down-sampling by the ganglion cells [4, 10], divisive normalization [12, 33], and intrinsic position uncertainty [28, 27, 16]). The solid curves in Fig. 3 show the predictions of the model observer after ML estimation of an overall efficiency parameter that scales all of the model's d' values. There are some systematic deviations from the predictions, but the theory quantitatively captures most of the variation in the data. Also, our theory for additive targets is a special case of this theory for occluding targets, and hence this theory also accounts for our earlier results on detection of additive targets in natural backgrounds [33]. This project gives us some confidence that we will be able to get a good handle on texture discrimination across the visual field.

C.1.3 Aim 3: Grouping and Segmentation of Natural Texture Patches

Aim 3 is directed at getting a better understanding of the functional mechanisms that group small texture regions into larger structures. Previous research on perceptual grouping has mostly focused on tasks with relatively simple stimuli designed to study a single kind of grouping principle: proximity, similarity, common motion, and good continuation [39]. These grouping principles were first revealed by Gestalt psychologists using cleverly designed images without reference to the functional role of these principles in normal vision. However, they all have a rational basis in the physics of natural environments and play an essential role in our ability to correctly interpret natural scenes. A useful paradigm for quantitatively studying perceptual grouping is to measure how grouping performance degrades with increasing stimulus noise. Classic examples involve varying line element orientation noise in studying good continuation [12, 18] and moving dot noise (level of coherence) in studying common motion [5, 32]. Here we propose to measure human ability to identify the dominant organization (rows or columns) in arrays of texture patches where the level of coherence is varied. So far, we have designed the stimuli and collected enough pilot data to know that the task produces systematic changes in performance.

C.2 Proposed Studies

C.2.1 Aim 1: Discrimination of Separated Natural Texture Regions

This aim is directed at measuring and modeling human ability to identify whether two separated texture patches are samples from the same or different natural texture, where the natural textures vary randomly on every trial and are drawn from a large enough set that there is no real opportunity for significant learning about the specific textures. In other words, we are interested in human ability to identify whether two patches contain the same content (stuff) even when that specific content is novel (not seen before). We believe this ability is a fundamental building block underlying human ability to group and segment natural scenes into regions likely to contain the same physically meaningful content. There has been little previous research directed at measuring and modeling this ability.

C.2.1.1 Texture content discrimination as a function of retinal eccentricity.

Same-different texture discrimination performance will be measured in the fovea (0°) and at three retinal eccentricities (1.6° , 5° , and 11.5°) picked to approximately correspond to the eccentricities where the spacing between ganglion-cell receptive fields is 2, 4, and 8 times that in the fovea [10]. The texture patches will be $1^\circ \times 1^\circ$, which corresponds to a typical size of receptive fields in the early visual cortical areas. On each trial the stimulus will be presented for 250 ms and the subject's task will be to report "same" or "different." In this first experiment, the 64×64 pixel texture patches will be randomly sampled from 60 larger (640×640) texture images from the classic Brodatz texture dataset, which were selected because they appear to be relatively dense and statistically stationary. No specific patch will appear more than once, but random samples from each of the 60 textures images will appear in 5 same and 5 different trials at each eccentricity, for the two configurations shown in Fig. 4 (a total of $600 \times 4 \times 2 = 4800$ trials). To reduce the possibility that the discrimination is based primarily on just the mean luminance and contrast differences, each patch will be normalized to have the same mean luminance and RMS contrast.

In the first configuration (Fig. 4A) the two texture patches will be at the same eccentricity. In the second configuration (Fig. 4B) one of the patches will always be in the fovea and the second patch will vary in eccentricity. The visual system often compares image patches at both similar and very different eccentricities, but it is unknown how human performance varies along this dimension. Formally, performance should be limited by the information in the most peripheral patch and thus performance may be similar in the two configurations. This would be the simplest result, but there are plausible models where this would not be the case (see modeling in the next section). There are many other texture databases available, and we will replicate this experiment with

a different set of textures in order to test generality. We plan to obtain measurements on at least 5 observers for the same set of stimuli presented in a different random order. These experiments will provide a rich dataset for testing models of texture discrimination. At a coarse level, the data can be summarized for each eccentricity by a same-different error matrix, where the diagonal elements are the number of errors for the same trials and the off-diagonal elements (above the diagonal) are the number of errors for each tested different pair. These matrices can be formed for the individual subjects and for the combined subjects. The models will make predictions for these error matrices and for the trial-by-trial responses to the specific individual stimuli. Some pilot data for one subject are shown in Fig. 4C and in Fig. 5.

To compare with Freeman et al. [14] and Lieber et al. [25], we will measure same-different discrimination of texture patches from their phase-scrambled versions, where again the luminance and contrast have been equated. However, it is worth noting that phase scrambling tends to Gaussianize the gray scale histogram and thus phase scrambling only preserves the second-order statistics. Thus, we may find that first-order statistics are able to explain some of the ability of the periphery to discriminate original and phase-scrambled patches.

C.2.1.2 Bayesian image-statistics models of texture content discrimination.

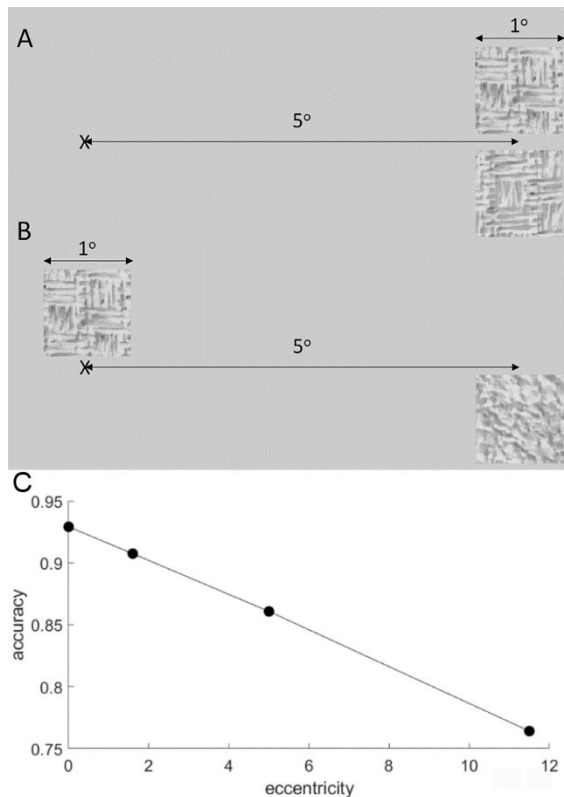


Fig. 4 Texture discrimination experiments. **A.** Both texture samples are at the same eccentricity (here the two samples are from the same natural texture). **B.** One sample is always in the fovea and the other sample varies in eccentricity (here the samples are from different natural textures). The task is to report “same” or “different”. **C.** Preliminary data for one subject.

log likelihood of same vs. different texture patches given this descriptive probability density. For example, the power at each spatial frequency is approximately exponentially distributed, consistent with the assumption that the patches are samples of Gaussian noise with arbitrary power spectra. The closed-form expression simultaneously estimates the two power spectra and the log likelihood ratio. This computation could be implemented heuristically by summing properly normalized complex-cell responses in the two 1° regions. However, our primary goal here will not be to explore biologically plausible implementations, but rather to characterize the stimulus information available for performing the task and to determine what components of that information underly human performance. Similarly, we derived an exact closed-form expression for the log

An important aim of the proposed research is to develop principled models of texture-content discrimination that can be tested against the experimental data. One class of models we will consider are Bayesian models based on specific sets of statistical properties of natural textures. For each statistical property, we determine/derive the log likelihood that the pair of samples of that property (from the two stimuli) are from the same vs. different textures. For example, each point in Fig. 5 represents a pair of log likelihoods: L_P and L_H , where L_P is the log likelihood that the power spectra are the same and L_H is the log likelihood that the gray-scale histograms are the same. The fine blue dots are for all the same trials and the fine orange dots are for all the different trials, where 10 same and 10 different trials were randomly sampled from every possible pairing of the 60 Brodatz textures (35,400 trials total). This figure shows that even with just these two principled log-likelihood estimates, it is possible to perform same-different discrimination at 93% correct with a single fixed decision bound (black curve). Given the way these textures were selected it is likely that performance with this specific decision bound would remain high for a very large number of arbitrarily selected natural textures. This encouraging result strongly suggests that relatively simple local image processing and simple grouping rules could go a long way toward supporting human ability to automatically group and segment images into physically meaningful regions, even for texture content that has never been experienced before.

Much of the reason for the success of this demonstration was due to deriving principled log likelihood measures. First, we find a one or two parameter family of probability densities that provide a good descriptive fit to the measured distributions across a wide range of textures. Then we derive an exact closed-form expression for the

likelihood of same vs. different gray-scale histograms assuming the histograms of the two patches are samples from arbitrary multinomial distributions. This expression simultaneously estimates the two multinomial distributions and the log likelihood ratio.

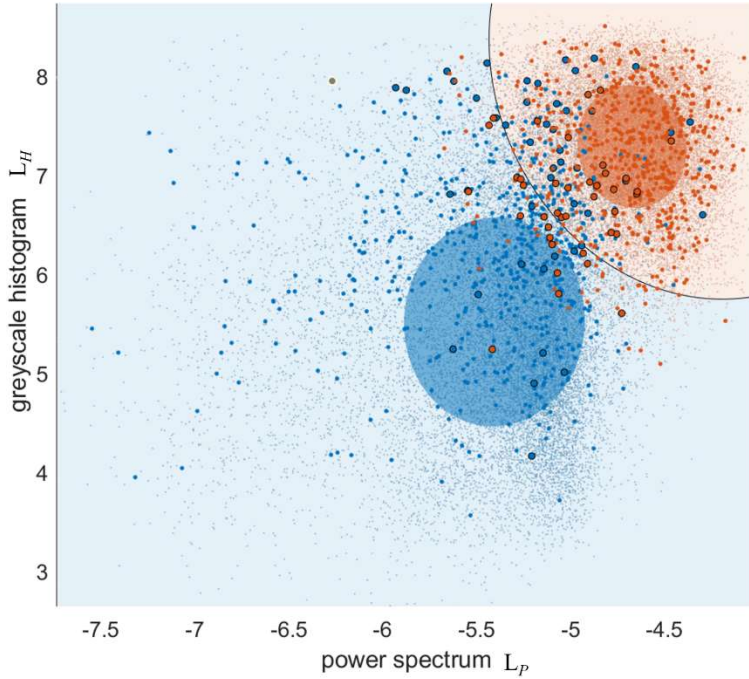


Fig. 5 Same-different texture discrimination of 1x1 deg patches based on natural scene statistics. A single decision bound (black curve) achieves 93% correct for random samples (smallest dots) from 60 Brodatz textures. The other dots are stimuli for corrects (medium dots) and errors (large dots) at 0° eccentricity in the pilot experiment (Fig. 4).

the effects on same-different texture discrimination. Once the texture patches are down sampled, they are compared using the same procedure as described above for the fovea. However, we allow a different decision bound to be learned for each retinal eccentricity. When the two patches are at different eccentricities, we assume that the one at the smaller eccentricity is effectively filtered and down sampled to match the eccentricity of the more eccentric patch. These are reasonable initial hypotheses given all the experiential and/or evolutionary time available to learn the differences between foveal and peripheral representations as well as the fixed decision boundaries. The experiments should provide a good test of these hypotheses.

C.2.1.3 CNNs and other models of natural texture content discrimination.

The other class of models we will consider are supervised and self-supervised convolutional neural networks (CNNs). Our initial supervised network is the relatively standard CNN architecture shown in Fig. 6. The input is a pair of 64x64 image patches stacked into a single block. The input is processed through three convolutional stages (containing increasing numbers of layers) and then two fully connected layers. In a first pass, we trained the network on texture patches randomly sampled from the 60 Brodatz textures. Although this CNN learned to perform well on independent test sets of random samples from the 60 textures (curve in Fig. 6), there was some

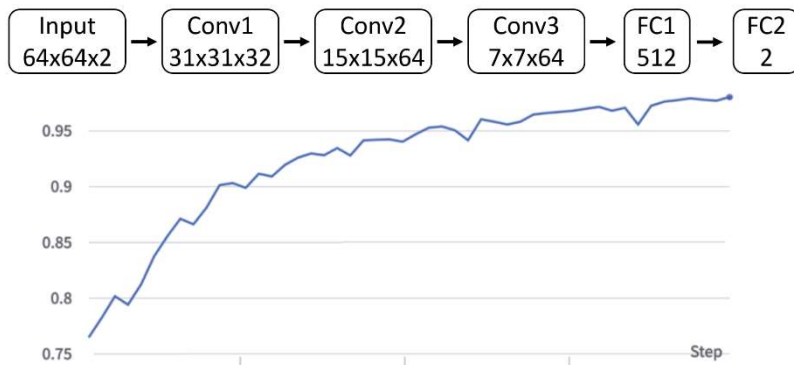


Fig. 6 CNN performance on same-different task as a function of training steps. Performance is for different random patches from the same set of 60 Brodatz textures used for training.

overfitting that we plan to eliminate by increasing the number of different types of texture in the training set.

To incorporate the effect of retinal location (eccentricity) into the theory, we filter and down-sample the texture patches based on ganglion-cell receptive-field shape and sampling density at the retinal locations of the texture patches. We have shown in the past that this does a reasonable job of predicting the effects of retinal eccentricity on detection in noise and natural backgrounds [4,41], although it remains to be seen how well it can predict

the effects on same-different texture discrimination. Once the texture patches are down sampled, they are compared using the same procedure as described above for the fovea. However, we allow a different decision bound to be learned for each retinal eccentricity. When the two patches are at different eccentricities, we assume that the one at the smaller eccentricity is effectively filtered and down sampled to match the eccentricity of the more eccentric patch. These are reasonable initial hypotheses given all the experiential and/or evolutionary time available to learn the differences between foveal and peripheral representations as well as the fixed decision boundaries. The experiments should provide a good test of these hypotheses.

As described in the background and preliminary studies (C.1.1), potentially more powerful, flexible, and biologically plausible models of texture-content discrimination can be developed using contrastive self-supervised learning. Specifically, simple discrimination of position and color can be used to bootstrap the learning of texture-content discrimination. The key assumption is that image patches that are very close together

and very similar in average color are likely to contain the same texture, whereas image patches that are well separated in space and color are likely to contain different textures. These differences in position and color will be used to create approximate ‘ground-truth’ labels for training CNNs. As mentioned earlier, CNNs are known to be highly robust to ground truth labeling errors [30].

We will train our first SSCNNs with patches sampled from our database of outdoor natural images (<https://natural-scenes.cps.utexas.edu/db.shtml>), but we will also train on other natural-image databases. Indeed, the advantage of this approach is that there is no need to know the ground truth during the training phase, so any set of images can be used. First, an image and then a patch from that image are selected at random. A “same” patch is then selected based on two parameters: a spatial distance radius in pixels s_d , and a color distance radius in CIELAB color space s_{lab} . If such a patch is not found, then a new starting image patch is randomly selected. After a “same” patch is found, a “different” patch is selected based on two more parameters, d_d and d_{lab} . Specifically, patches are randomly sampled at a spatial distance greater than d_d until a patch with chromatic distance greater than d_{lab} is found. These position and color distance parameters are something evolution could find, but we will simply vary them until we get subjectively reasonable patches that achieve good performance on one or more ground-truth texture databases. Our initial implementation of this sampling scheme seems to produce a large percentage of subjectively plausible labels. The randomly selected same and different patches will be converted to gray scale (e.g., like the Brodatz textures) before training the network. Again, we will start with the network architecture shown in Fig. 6.

Although there is no limit to the amount of training data available for training SSCNNs, our goal is not to develop practical applications, but to develop models with potentially interpretable/analyzable convolutional layers [45,46]. Thus, we will focus on relatively small networks like the one in Fig. 6. After a SSCNN is trained, we will test it on ground-truth labeled databases, such as the Brodatz textures, to see how its absolute performance compares with that of the same network trained in a supervised fashion with correctly labeled training patches.

The network models will be trained separately for the four different eccentricities in the texture discrimination experiments. To do this, the input patches will first be filtered by the average optics of the human eye [43] and then down sampled by an array of ganglion cells consistent with the anatomy of the human retina [10,4]. Separate training is justified because there is just as much evolutionary and experiential time available to the fovea and periphery. This kind of training is appropriate for predicting performance in the version of the same-different task in Fig. 4A, but not for the task in Fig. 4B where the pair of textures being compared are at different eccentricities.

To make predictions for the task in Fig. 4B, we will take two different approaches. One approach will be the same we propose to use for the Bayesian image-statistics models. We assume that the visual system has learned (through evolution and experience) what happens to the internal representation of image patches when they are moved from one location to another on the retina. Importantly, a reliable training signal is available almost every time we move our eyes. Specifically, when we move our eyes the probability that the image patch from a particular world location changed is very small, and thus one can be almost certain that any changes in the representation of that image patch are due the change in retinal location (i.e., the two representations can be labeled as “same”). If we assume that the effect of retinal location is precisely known through evolution and experience, then we can generate predictions for the task in Fig. 4B simply by blurring and down sampling the representation of the patch at 0° and applying the network learned for the task in Fig. 4A. The second approach will be to train a self-supervised network to predict the peripheral representation from the foveal representation and use that predicted representation as the input to the network learned for the task in Fig. 4A.

As mentioned earlier, we will compare the Bayesian image statistics models, the SSCNN models, and several other models (FRF and P-S) against the experimental data. In all cases we apply the same optics and retinal down sampling to the input image patches. None of the models are trained to predict human performance and they all generate a continuous decision variable, and thus they can be evaluated by comparing goodness of fit after minimizing errors with only a single overall efficiency scalar.

C.2.2 Aim 2: Discrimination of Adjacent Natural Texture Regions

Texture grouping and segmentation is based on both the discrimination of texture content and the detection of texture boundaries. In this aim, we focus on the task of detecting texture boundaries both when the texture content is the same in both regions and when it is different. The approach will closely parallel the approach described above for discrimination of texture content. First, we will measure human texture boundary detection

performance for a wide range of natural textures, where the texture type is the same on both sides of the boundary (maximally camouflaged targets). Second, we will measure texture boundary detection when the textures are different on the two sides of the boundary. Finally, we will develop, and test, Bayesian image-statistics models and CNN based models which we will compare to the human boundary-detection data.

C.2.2.1 Discrimination of adjacent regions containing the same natural texture

As described in the progress report, we already completed multiple parametric experiments measuring detection performance for maximally camouflaged targets (targets with the same texture as the background). These experiments primarily measured performance under highly controlled conditions where the type of texture was blocked and where the number of different types of texture was relatively small. Like the discrimination of texture content described in Aim 1, under natural conditions we often must detect boundaries between textures that randomly vary over a wide range and that we may never have seen before. Thus, we propose experiments to measure detection of texture boundaries that parallel those proposed to measure texture content discrimination. In the first experiment, the stimuli on each trial will be like those in Fig. 4A (see also Fig. 1B), except instead of two texture patches there will either be a single vertical $1^\circ \times 2^\circ$ patch of texture or two abutting $1^\circ \times 1^\circ$ patches sampled from the same type of texture. The subjects' task will be to report whether there is a border present (one larger patch or two smaller patches). Again, performance will be measured in the fovea (0°) and at three retinal eccentricities (1.6° , 5° , and 11.5°) for randomly sampled patches from the 60 Brodatz textures tested in the texture content discrimination experiment. In the second experiment, we will replicate the first experiment with textures selected from a different dataset to test the generality of the results and models.

C.2.2.2 Discrimination of adjacent regions containing the same or different natural texture

The third experiment will be essentially the same as the experiments above except that when there are two texture patches, they will be patches sampled from different textures. The detection of the two regions can now be mediated by both the texture boundary cues and the texture content cues. Comparison of the results from this experiment with those for texture content alone (Fig. 1A) should reveal when and how much the boundary cues contribute to the discrimination of adjacent texture regions.

C.2.2.3 Bayesian image-statistics models of texture boundary detection

The modeling of texture boundary detection will use the same approaches used in modeling texture content discrimination. We will first consider the case where the texture boundary is between two samples of the same texture (the maximum camouflage case, C.2.2.1). As described in the progress report, we will first characterize the shape of the statistical distribution for multiple natural image properties within the region straddling the potential texture boundary. Specifically, for each texture type we will sample a large number of texture patches. Then for each sample region, we filter with the optics of the eye and extract contours. We then measure the properties of the contours within the straddle region. These properties include the total number of contour pixels, the length of each contour, and the edge energy of each contour at five different scales (other properties we have examined so far did not contain significant additional information). From these measurements we compute histograms, which we analyze to determine a one- or two-parameter distribution shape. Once these shape families are known, it is possible to estimate the one or two parameter values and then the log likelihood ratio of contour present or absent, for each property on each trial. A large sample of these log likelihood ratios for a wide range of textures will be used to determine a single fixed optimal decision bound for responding whether the texture boundary is present or absent. As described earlier, this will be done for each retinal eccentricity in the experiment. We will then test alternative models by making trial-by-trial predictions for the experiments in C.2.2.1. We will then consider the case where the texture boundary is between two samples from different textures (different trials will be abutting samples from different textures, C.2.2.2). In this case, the models will contain both texture content and texture boundary cues.

C.2.2.4 CNNs and other models of natural texture boundary detection

Analogous to Aim 2, we will consider supervised and self-supervised CNNs. In each case, we will consider two versions, one that is trained on the straddle regions and one that is trained on the whole image. CNNs trained on the whole images might be particularly useful because they should reveal the relative amount of discrimination information contained in features near and away from the border. To compare with earlier work, we will also test the FRF models.

C.2.3 Aim 3: Grouping and Segmentation of Natural Texture Regions

Our ability to group and segment images into physically meaningful regions builds upon our fundamental ability to discriminate texture content and to detect texture boundaries. Using the models of texture discrimination developed in Aims 1 and 2 as a foundation, we will develop and test theories of texture grouping and segmentation that incorporate multiple grouping principles, three of which are well-known and one of which is new. The three well-known principles are the Gestalt principles of grouping by proximity, similarity, and good continuation (co-linearity) [39]. The new principle is a simple recurrent operation we call “mutual similarity” grouping (see below). In the human visual system, these principles are likely to operate in a hierarchical fashion to obtain meaningful regions (and subregions) at multiple scales. Here, however, we focus on a single level of grouping, which is sufficient to demonstrate the four principles and to generate predictions for the proposed experiments.

C.2.3.1 Texture grouping and segmentation experiments

A simple “rows or columns” (RC) task will be used to measure human texture grouping and segmentation performance and to test the proposed theories. Typical stimuli in the task are illustrated in Fig. 7. On each trial, a stimulus will contain randomly selected patches from two randomly selected types of texture, where the dominant organization of the patches is either in rows or in columns. The stimulus will always contain an equal number (18) of each type of texture. The coherence of the stimulus will be varied by varying the proportion of patches in the dominant organization. When the proportion is high (e.g., $p = 0.89$ in Fig. 7) it is relatively easy to judge whether the dominant organization is rows or columns. As the proportion nears 0.5, performance must approach chance (right side of Fig. 7). In the first experiment, the texture patches will be sampled from the same 60 Brodatz textures used in the first texture discrimination experiment (C.2.1.1). There will be four coherence levels (p) selected so that performance ranges from near chance to near perfect. The texture types will be sampled without replacement so that each texture appears twice in a 60-trial block. There will be 8 blocks in a session, 4 with the coherence decreasing followed by 4 with coherence increasing, and 6 sessions in the experiment (total trials = 2,880). Four to five Ss will be run on the same set of stimuli, with potentially more Ss depending on the magnitude of individual differences.

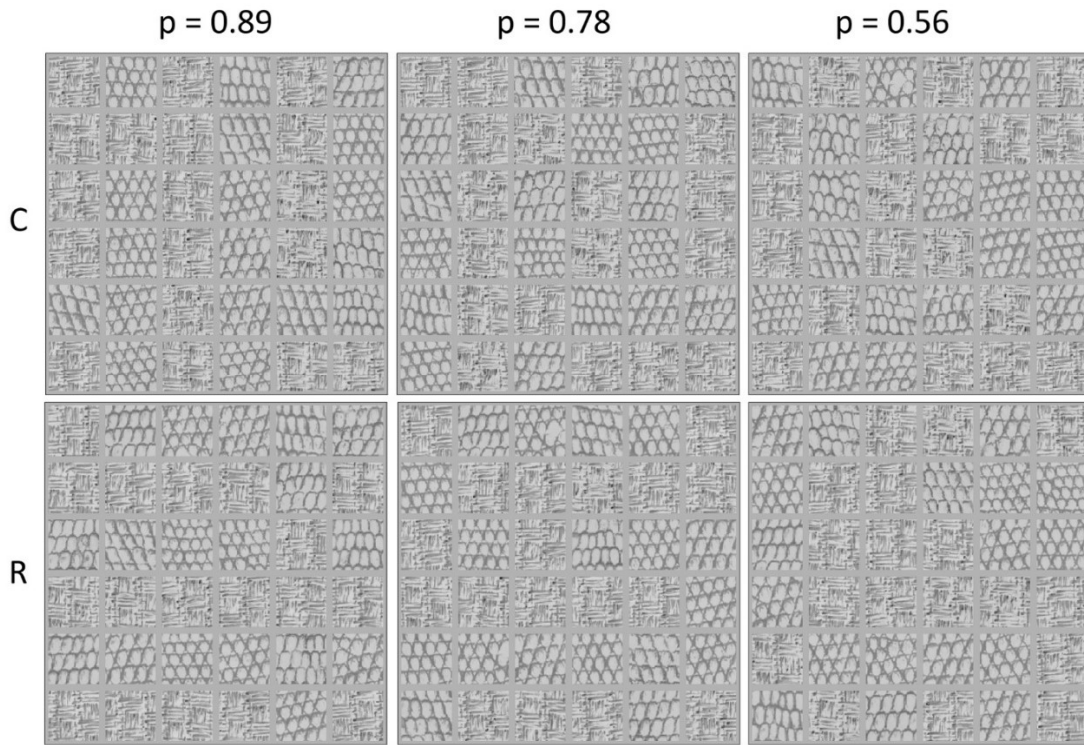


Fig. 7 Texture grouping experiment. In upper panels, the dominant organization is columns (C), lower panels rows (R). p = coherence.

Although the RC task is relatively simple, human performance should depend on all four of the grouping principles considered in the theory of texture grouping and segmentation mentioned above and described below. Furthermore, because all 2,880 stimuli contain different random samples of natural texture, the trial x trial

predictions generated by the models should provide a strong test of their validity. Once the specific models have been developed and tested against the data from this experiment, a second experiment will be run with natural textures drawn from a different dataset. A strong test of the models developed for the first experiment will be their ability to generalize to a different natural texture database.

C.2.3.2 Theory of texture grouping and segmentation

The proposed theory of texture grouping and segmentation is based on the four principles mentioned above. Consider first the principles of similarity and mutual similarity grouping. Fig. 8 illustrates the effect of these two grouping principles based on the same-different discrimination decision bound in Fig. 5. Specifically, Fig. 8A shows 16 large randomly sampled patches of Brodatz texture, which we divide into 16 $1^\circ \times 1^\circ$ patches. Fig. 8B shows groupings obtained by similarity grouping alone (no proximity or other cues). Each of the 256 small patches was compared with every other patch and a link was assigned between the patches if they were categorized as “same”. A unique group label (color in the figure) was assigned to each set of patches that were linked together. The goodness of the grouping is assessed with two scores: the within-region accuracy a_w and the across-region accuracy a_A . To calculate these scores, we first determine the dominant label (color) for each of the 16 different ground-truth texture regions. The within-region accuracy is defined to be the fraction of patches within the region that have the dominate label (averaged over all the ground-truth regions). The across-region accuracy is defined to be the fraction of ground-truth regions having a unique dominant label.

We believe mutual similarity grouping is a new principle. The idea is that a pair of linked patches that are both linked to the same set of other patches are more likely to be the same texture and therefore should be more tightly linked. On the other hand, if the two patches are linked to different sets of other patches, then they are outliers to each other and are less likely to be the same texture. Thus, when the mutual similarity between two patches is high the link strength between them increases, otherwise it decreases. This operation is inherently recurrent: first the similarity links are formed and then there is feedback to adjust the link strengths based on the mutual links. Fig. 8C shows the effect of applying this recurrent operation to the groups in Fig. 8B. As can be seen, both grouping scores increase substantially.

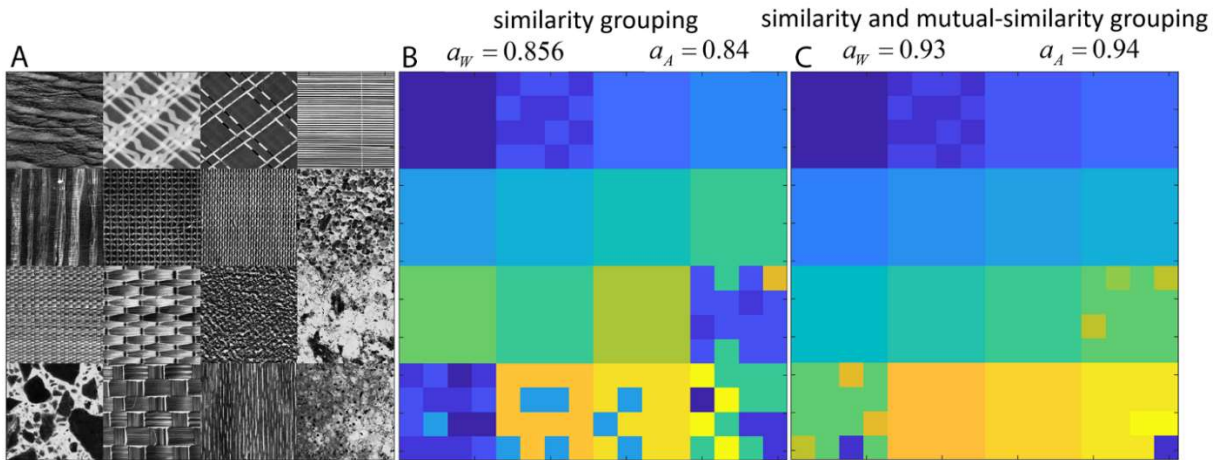


Fig. 8 Demonstration of similarity and mutual-similarity grouping.

To introduce the role of the other two grouping principles, consider the predictions for RC tasks like that in Fig 7. First, perform similarity and mutual similarity grouping to obtain initial links between all patches. Second, only keep those similarity links that are sufficiently proximal. For simplicity, we assume that only abutting texture patches are sufficiently proximal (although this can be relaxed). Third combine linked patches that are co-linear to obtain the final set of groups. Finally, respond “rows” or “columns” based on the distribution of group lengths: more large-horizontal groups imply the organization is rows; more large-vertical groups imply the organization is columns. We have implemented a simple version of this model and it performs well in the proposed RC task. Although the model is largely based on the statistics of natural textures, there are several free parameters and there are versions of the model that differ in certain details (e.g., the Bayesian or SSCNN model of same-different texture discrimination). Once the first experiment and modeling are completed, we will determine which versions of the theory best explain the data from the RC task. We will then test how well the theory generalizes by generating predictions for the second experiment using natural textures from a different dataset.